PDF → AS 1100 drawing pipeline

Read the true geometry, then re-draft it to AS 1100. Rev C, 13/11/2026

MVP step · supporting / data · later

B badge = revision that added or changed the box

Read the drawing

1 Triage
Vector or raster; set scale in mm

2 Vector extract
pdfplumber paths + text → JSON sketch

3 Raster vectorize
SegFormer + DeepLSD + arc fits

4 Read annotations C
eDOCr2 / VLM: dims, GD&T, title block (local models only)
Every dimension read is checked against step 6; disagreement → a person

solved geometry checks each dimension read

5 Understand
Graph model: roles, links, constraints, views

5b Components
Spot, match and size standard parts from the catalogue

6 Param. solve
FreeCAD / planegcs; OCR dims drive; conflicts flagged

templates

Re-draft to AS 1100

7 Geometry clean-up
OCCT: trim, join, close profiles; check vs NAS values

8 View normalisation C
Swap / realign views that exist. Never invent a projection the source did not show

9 Re-draft to AS 1100
Model + drafting layer → DXF: dims, blocks, parts list, balloons

9b Title block, no., rev C
Client's own rev system kept (BHP's is BHP's). File name = drawing no.; file rev = title block rev

10 Check & review 1st
Overlay diff, DOF = 0. Edits → model (6) or drafting layer (9)

11 Human sign-off C
A drafter checks the overlay report and every flagged item. Nothing issues unchecked.

Outputs
DXF · PDF · BOM (parts list, CSV/XLSX) · JSON/FCStd · STEP

known dims / scale check

style rules

corrections

Supporting tracks

NAS reference catalogue B
Known sizes, scales, part dims from past jobs. SQLite, read-only. Final issued rev only. Hints, not truth. No client names. Crawler: todo

0 AS 1100 profile
.101, .201 + HB 47 as a DXF template: layers, linetypes, dim style, sheets, projection symbol

Synthetic data
build123d/OCCT parts → HLR views → AS 1100 / ISO / ASME → PDF/PNG. Trains steps 3–5

Parts catalogue
Parametric templates + BOM data: fasteners, flanges, bearings, sections

Testing

Overlay check (built) B
Render result over source; per-entity scores; numbers must match. overlay-check/ in the repo; self-test passes

Synthetic round trip
DXF → render → pipeline → compare, at rising degradation

Frozen real test set B
Past jobs from the NAS, redrawn by a drafter. Never used for training.

Metrics + CI gate
F1, CER, DOF; dimension satisfaction, fix count. AS 1100 linter, DXF opens in ODA / LibreCAD

Build order

0. Overlay check (step 10 diff) + frozen test set of past jobs. Overlay check built.
1. MVP: clean vector PDFs, rules only (terracotta steps: 0, 2, 6, 7, 9, 9b, 10, 11, outputs)
2. Synthetic data generator
3. Graph model replaces rules in step 5
4. Parts catalogue + step 5b (~10 families)
5. Raster path (step 3)
6. GD&T and multi-view

Decisions still open

Hosted AI vision model for client drawings? Assume NO: local models only until decided.
Three BHP revision / file-naming questions: not yet received; add when available.
NAS: where the crawler runs, what is on it, and whether any content is confidential.
Licences: PyMuPDF is AGPL (the overlay check uses pdfplumber instead). Check the eDOCr2, DeepLSD and SegFormer weights before they go into the worker.

PDF → AS 1100 DRAWING PIPELINE — **REV C**

DATE 13/11/2026 — SCALE N.T.S — DWG NO. DC-AUTO-PIPE

REV	DATE	WHAT CHANGED
A	13/11/2026	First diagram, as received: steps 0 to 10, build order, testing row.
B	13/11/2026	Overlay check first, frozen real test set, local models only, NAS reference catalogue, decisions still open.
C	13/11/2026	Reviewer items: human sign-off (11), title block / drawing no. / revision step (9b), step 4 cross-check, step 8 limit, licences in open decisions. Title block added.